The Standard Subgraph Classification Datasets: Four Real and Four Synthetic

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What these datasets are

Subgraph classification has a standard test set. It came with SubGNN (Alsentzer et al., NeurIPS 2020), and almost every later method reports on it: GLASS (2022), S2N (2024), SSNP (2023) and GPEN (2025). It has four real datasets. In each one there is a single large graph, a list of subgraphs inside it, and one label per subgraph. SubGNN also made four synthetic datasets: Density, Cut-Ratio, Coreness and Component. They have their own section below.

All four real datasets are already on this PC, in `data/raw/subgnn`. SubGNN did not release the synthetic datasets, only code to make them. GLASS published the files it used, and we downloaded those into `data/raw/subgnn_synthetic`. Every number below was measured from these files, unless it says it comes from a paper.

Dataset	Node	Subgraph (one color)	Label of the subgraph	Classes
Pregnancy (ours)	a visit	a patient's visits	had the outcome	2
PPI-BP	a protein	the proteins of one biological process	the kind of process	6
HPO-Metab	a symptom or a gene	the symptoms of one rare disease	the kind of metabolic disease	6
HPO-Neuro	a symptom or a gene	the symptoms of one rare disease	the kinds of neurological disease	10, several each
EM-User	a workout	a user's workouts	the user's gender	2

Table 1: How each dataset is split into subgraphs and what each subgraph is labelled with.

Each dataset

PPI-BP Human proteins, with an edge when two proteins are known to interact. A subgraph is the set of proteins that take part in one biological process, taken from the Gene Ontology gene sets. The label is the broad group of that process: metabolism, development, signal transduction, stress and death, cell organization, or transport.

HPO-Metab A knowledge graph of symptoms (from the Human Phenotype Ontology) and genes. A subgraph is the list of symptoms of one rare disease. The label is one of six kinds of metabolic disease, for example lipid or lysosomal. Each class has exactly 400 subgraphs.

HPO-Neuro The same graph as HPO-Metab, with neurological diseases instead. A disease can be in more than one of the ten classes, so the label is a set of classes.

EM-User Workouts from the fitness app Endomondo. A subgraph is the workouts of one user, and the label is the user's gender. It is the only binary dataset, and it is balanced: 162 women and 162 men.

Size

	Nodes	Edges	Avg. degree	Subgraphs	Avg. size	Median	Smallest–largest
Pregnancy (ours)	61,046	972,557	31.9	6,590	9.3	5	2–221
PPI-BP	21,521	338,625	31.5	1,591	10.2	7	1–160
HPO-Metab	14,587	3,238,174	444.0	2,400	14.4	13	1–50
HPO-Neuro	14,587	3,238,174	444.0	4,000	14.8	14	2–55
EM-User	57,333	4,573,417	159.5	324	155.4	134	3–499

Table 2: Size of the graphs and of the subgraphs. Size is the number of nodes in a subgraph. Edges are counted once, without self-loops.

How the subgraphs sit in the graph

	Of all nodes, in no subgraph	Of nodes in a subgraph, in two+	Pieces per subgraph	One piece	Edges inside a subgraph	Same-label neighbours (chance)
Pregnancy (ours)	0%	0%	8.8 of 9.3	–	0.4%	83.0% (81.7%)
PPI-BP	67.4%	47.8%	7.0 of 10.2	2.1%	2.3%	27.5% (19.4%)
HPO-Metab	82.5%	78.2%	1.6 of 14.4	54.8%	2.9%	21.8% (17.0%)
HPO-Neuro	76.3%	78.2%	1.5 of 14.8	59.3%	5.1%	–
EM-User	87.6%	68.0%	30.6 of 155.4	0%	1.1%	49.6% (50.0%)

Table 3: The first two columns have different bases: the first is out of all nodes, the second only out of the nodes that are in some subgraph. *Pieces*: take the nodes of one subgraph, keep only the edges between them, and count the separate groups that are left. A node with no edge to the rest of its subgraph is a piece on its own. So "7.0 of 10.2" means an average subgraph of 10.2 nodes splits into 7 groups that do not touch each other. *One piece* is the share of subgraphs that do not split at all. *Edges inside a subgraph* is the share of all edges in the graph whose two ends are in the same subgraph. *Same-label neighbours* looks at edges between two nodes of different subgraphs, using only nodes that belong to one subgraph, and asks how often both subgraphs have the same label; *chance* is what random edges would give. It is left out for HPO-Neuro, where a subgraph has several labels.

Most nodes belong to no subgraph. Between 67% and 88% of the nodes are only background. In our data every visit belongs to a patient.

Subgraphs share nodes. About half of the PPI-BP proteins that are in a subgraph are in two or more, and about three quarters in HPO and EM-User. In our data a visit belongs to exactly one patient.

PPI-BP and EM-User fall apart like ours. A PPI-BP subgraph of 10.2 proteins splits into 7.0 pieces, and only 2% are in one piece. An EM-User subgraph splits into 30.6 pieces. The HPO subgraphs are mostly in one piece.

Neighbours carry some label signal, except in EM-User. In PPI-BP, neighbouring subgraphs share a label 27.5% of the time, against 19.4% by chance. In EM-User the two numbers are the same, as in our data.

Size says little about the label. In EM-User women and men have about the same number of workouts (152 and 158). In PPI-BP, signal transduction subgraphs are the largest (13.9 proteins on average) and transport the smallest (8.6), so size helps a little. In HPO-Metab lysosomal diseases are the largest (18.2 against 11.1 for carbohydrate).

The synthetic datasets

Each one is a single made-up graph with subgraphs planted in it. The label of a subgraph is a property of its shape, fixed when the dataset was made. The nodes have no features, so a model can only use the structure.

Density The label is the share of a subgraph's possible edges that are present: low, medium or high. In the files the three ranges do not overlap: up to 0.074, from 0.079 to 0.168, and from 0.174.

Cut-Ratio The label is how many edges leave the subgraph, out of how many could. Of the edges that touch a subgraph, 77%, 80% and 83% leave it on average in the three classes. The subgraphs themselves are almost complete: 95% of their possible edges are present.

Coreness The label is how deep in the core of the graph the subgraph sits, in three classes. The average core number of its nodes is 19.4, 25.1 and 30.7 in the three classes, but each class covers almost the whole range, from about 3 to 43.

Component The label says whether the subgraph is one piece or several. All pieces, in both classes, have exactly 15 nodes. One class is a single piece of 15 nodes. The other is 5 to 10 pieces that do not touch each other, so 75 to 150 nodes.

	Nodes	Edges	Avg. degree	Subgraphs	Classes	Avg. size	Median	Smallest–largest
Density	4,998	29,962	12.0	250	3	20.0	20	20–20
Cut-Ratio	5,000	84,295	33.7	250	3	20.0	20	20–20
Coreness	5,000	119,205	47.7	221	3	20.0	20	20–20
Component	17,260	91,619	10.6	250	2	65.0	15	15–150

Table 4: Size of the synthetic graphs and of their subgraphs, counted as in Table 2.

	Of all nodes, in no subgraph	Of nodes in a subgraph, in two+	Pieces per subgraph	One piece	Edges inside a subgraph	Same-label neighbours (chance)
Density	55.0%	40.5%	8.1 of 20.0	3.2%	15.8%	32.9% (33.5%)
Cut-Ratio	37.5%	42.1%	1.0 of 20.0	100%	53.4%	31.4% (33.5%)
Coreness	11.6%	0%	1.0 of 20.0	100%	8.0%	38.6% (39.1%)
Component	5.8%	0%	4.3 of 65.0	50.4%	93.4%	82.6% (84.1%)

Table 5: The same columns as Table 3. Here *chance* is the sum of the squared label shares among the ends of the edges counted.

Size says nothing, or everything. In Density, Cut-Ratio and Coreness every subgraph has 20 nodes. In Component size alone gives the label: 15 nodes for one class, 75 or more for the other.

Neighbours carry no label signal. In all four, neighbouring subgraphs share a label as often as chance or a little less, as in our data.

Only Density falls apart like ours. A Density subgraph of 20 nodes splits into 8.1 pieces. Cut-Ratio and Coreness subgraphs are always one piece.

Coreness and Component do not share nodes. No node is in two subgraphs, as in our data. In Density and Cut-Ratio about 40% of the nodes in a subgraph are in two or more.

The test sets are small. The GLASS code draws a new random split in every run: half for training, a quarter for validation and a quarter for testing. That leaves 63 test subgraphs (55 in Coreness), so one subgraph more or less right moves the score by about 1.6 points.

Published results

	PPI-BP	HPO-Metab	HPO-Neuro	EM-User
SubGNN (2020)	59.9 ± 2.4	53.7 ± 2.3	63.2 ± 1.0	81.4 ± 4.6
GLASS (2022)	61.9 ± 0.7	61.4 ± 0.5	68.5 ± 0.5	88.8 ± 0.6
S2N, best version (2024)	64.3 ± 1.8	63.9 ± 1.7	68.6 ± 0.8	89.0 ± 1.6

Table 6: Micro-F1 on the test set, mean and spread over 10 runs, from Table 2 of the S2N paper (Kim and Oh, ICML 2024). For the binary EM-User this is the accuracy. The papers do not report AUC, the metric we use.

Scores differ a lot between datasets. The best methods reach 89% on EM-User but only 64% on PPI-BP and HPO-Metab, where there are six classes.

	Density	Cut-Ratio	Coreness	Component
SubGNN (2020)	91.9 ± 1.6	62.9 ± 3.9	65.9 ± 9.2	95.8 ± 9.8
GLASS (2022)	93.0 ± 0.9	93.5 ± 0.6	84.0 ± 0.9	100.0 ± 0.0
S2N+A (2024)	93.2 ± 2.6	56.0 ± 0.0	85.7 ± 5.8	100.0 ± 0.0
S2N+A with RWPE (2024)	93.6 ± 2.0	89.2 ± 2.6	72.6 ± 6.2	100.0 ± 0.0

Table 7: Micro-F1 on the synthetic test sets, mean and spread over 10 runs, from Table 3 of the S2N paper. The SubGNN and GLASS rows are copied there from their own papers. S2N+A is the GCNII version; RWPE adds random-walk position features to the nodes.

Component is solved, Cut-Ratio is not. Every method is at or near 100% on Component, where size gives the answer. Cut-Ratio spreads the methods most, from 56% to 93.5%.